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N + - Network Fundamentals & Building Networks.

Ethernet standard and specification.

TASK 1.

❖ Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

➤ RJ-45 Connector Pin Layout:

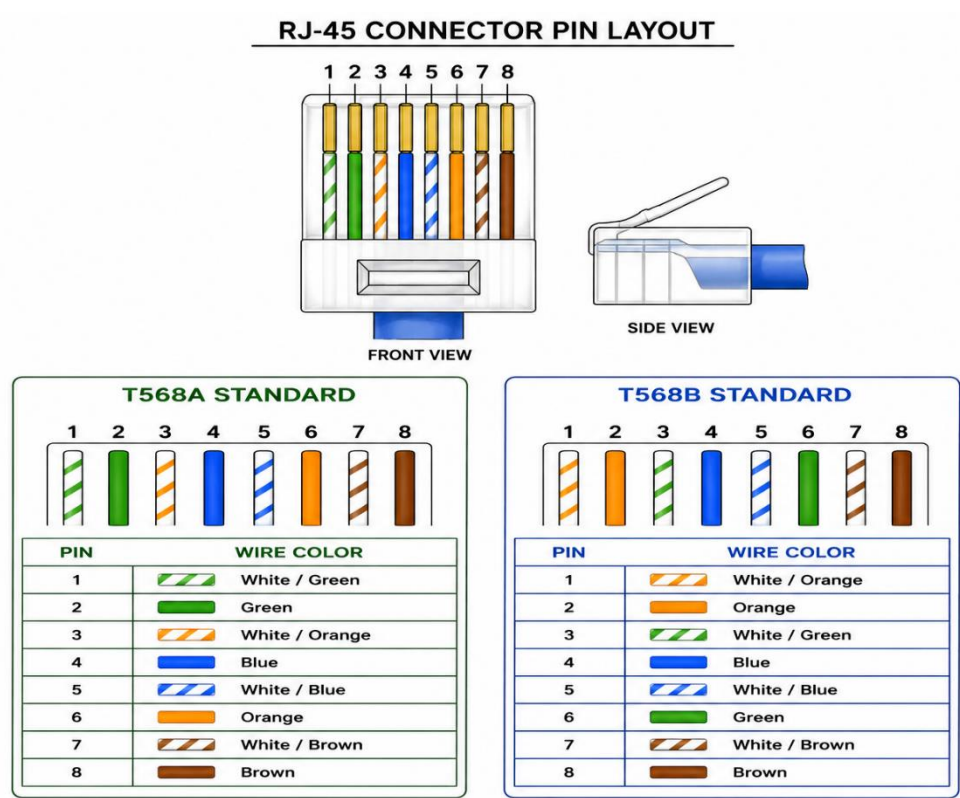


Figure : RJ-45 Connector Pin Layout and T568A/T568B Wiring Standards.

⌘ T568A Color Code:

PIN	WIRE COLOR
1	White/Green
2	Green
3	White/Orange

4	Blue
5	White/Blue
6	Orange
7	White/Brown
8	Brown

⌘ **T568B Color Code:**

PIN	WIRE COLOR
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

TASK 2.

- ❖ Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

➤ Comparison of Ethernet Standards

<u>Ethernet Standard</u>	<u>Speed</u>	<u>Cable Type</u>	<u>Maximum Cable Length</u>	<u>Connector Type</u>
10BASE-T	10 Mbps	Cat3 or better UTP	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better UTP	100 m	RJ-45
1000BASE-T	1 Gbps	Cat5e or better UTP	100 m	RJ-45

TASK 3.

- ❖ Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why, Think about how transmit and receive pairs are wired in each cable type.

➤ Reason:

A **crossover cable** is used to connect two laptops directly because both devices are similar network devices. It crosses the **Transmit (TX)** and **Receive (RX)** pairs, so the TX signals from one laptop reach the RX pins of the other laptop.

In comparison, a **straight-through cable** keeps the TX and RX pairs in the same order and is mainly used to connect different types of devices, such as a laptop to a switch.

Therefore, a crossover cable is the appropriate choice for a direct laptop-to-laptop Ethernet connection.

TASK 4.

❖ **List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.**

➤ **Real-World Devices Using Ethernet:**

Device	Where It Is Used	Likely Ethernet Standard
Wi-Fi Router	Home / College	1000BASE-T (Gigabit Ethernet)
Desktop Computer	College Computer Lab	1000BASE-T (Gigabit Ethernet)
Network Printer	College Office	100BASE-TX (Fast Ethernet)